

A red and green helicopter is shown in flight against a cloudy sky. The helicopter has "Gama Aviation" written on its side. The tail section features the text "Amdro Pymppwr. Dros Cymru. Serving Wales. Saving Lives." and the registration "G-LOYW". A green semi-transparent box is overlaid on the lower left, containing the title "Helicopter Emergency Rescue" and the text "GDP 2025".

Helicopter Emergency Rescue

GDP 2025

The Perfect Air Ambulance

Speed

- Start up
 - Call to lift
- Transit speed
 - Fast as possible
 - Efficient
- Turnaround
 - Refuel speed



The Perfect Air Ambulance

Size – small but big

- D-value
- Downwash
- Fuel burn
- Maintenance

Interior space

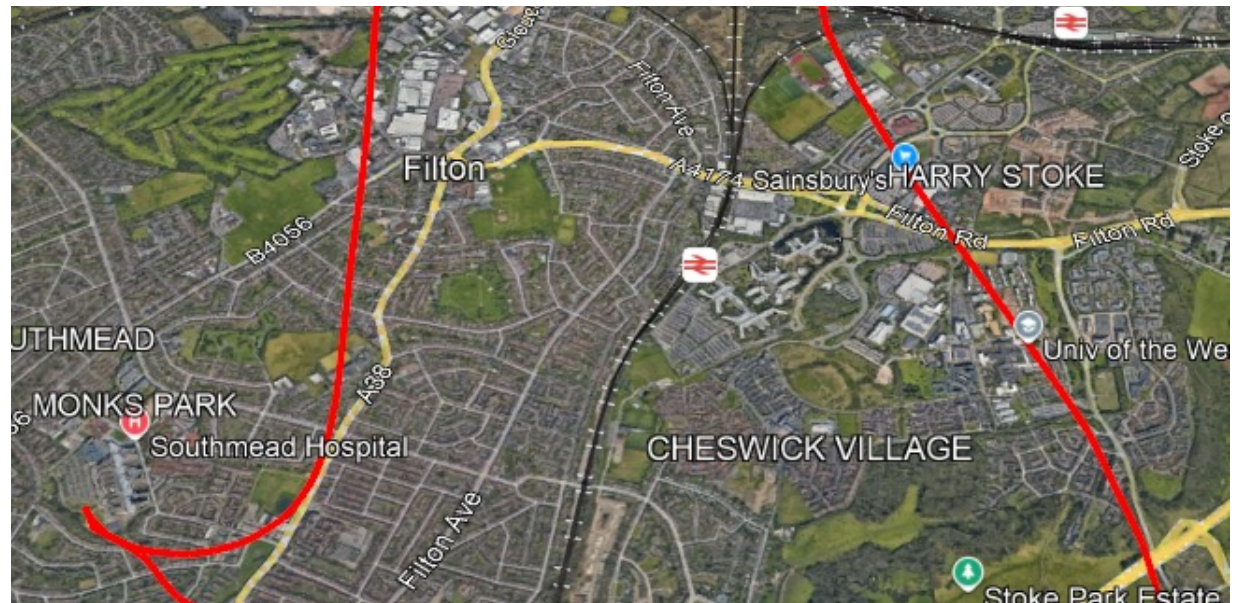
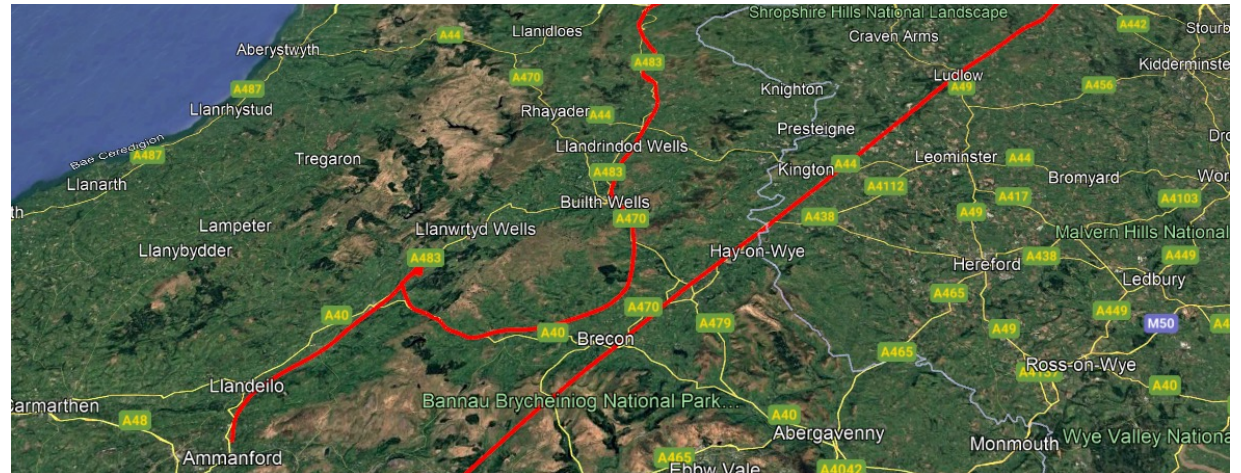
- Pilot
- 2 medics
- Patient
- Trainee/observer/checking
- Equipment
- Access



The Perfect Air Ambulance

Range

- Highly location dependent
- Short range is also limiting – payload!

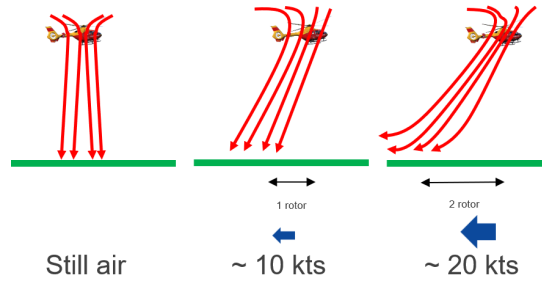


The Perfect Air Ambulance

Payload

- People (average 90 kg with kit each)
 - Pilot
 - Paramedic
 - Doctor
 - Patient
 - Extra person
- Fuel
 - Minimum 1 hour 15 min plus reserve
- Role fit (eg EC135)
 - 125 kg (not including seats) for fixed fittings
 - 117 kg for carry on equipment
- Total – 1140 kg – 740 kg without fuel





The Perfect Air Ambulance

Safety

- Twin – failure tolerant
- Automation
 - Reduce workload
 - Automated take off
 - IFR (Including weather radar)
 - PINS (Point in Space Approach)
 - NVIS (Night Vision Imaging System)
- Downwash
 - Minimise
 - Predictable

The Perfect Air Ambulance

- Availability
 - Maintainable – Spare aircraft available (target 98% availability)
 - IFR – Capable of operating in known icing conditions
 - Temperature tolerant – Hot and High, Cold and Low
 - Wind – 50 kt without limitation
 - Training – Flight Simulator availability, Instructor availability



Most common HEMS helicopters





Roles



FATT01 – Project coordinators

Academic advisor: Dr Ribera Vicent

- Project coordination & information management
- Cost estimation and market analysis
- Conceptual trade off studies & design
- Compliance with all relevant design standards and regulations

FATT02 – Aerodynamics

Academic advisor: Dr Steiros

- Rotor(s) design
- Design of lifting surfaces
- Airfoil selection/design/optimizations
- Fuselage and empennage aerodynamic design.
- If present, tail rotor design.

Roles

FATT03 – Structural Design & CAD

Academic advisor: Dr Shamsuddin

- Prediction of design loading conditions
- Rotor structural design and dynamic blade behaviour
- Selection of aircraft structural layout and sizing of major components
- Material and manufacturing method selection
- Landing gear design
- Hub design
- CAD modeling

FATT04 –Systems & Propulsion

Academic advisor: Dr Fasel, Dr Seifkar

- Powerplant type selection & integration
- Powerplant sizing and performance prediction
- Transmission considerations
- Vehicle sub-systems selection/sizing, layout and integration
- Sensors and avionics
- Weight & Balance Prediction
- Medical and mission specific equipment

Roles

FATT05 – Flight Mechanics, Performance & Control
Academic advisor: Dr Armanini, Dr Ribera Vicent
• Mission definition and operations • Vehicle static and dynamic stability prediction • Response to control inputs • Design of flight control system • Aircraft performance prediction



Courtesy of George Williams, former RAF test pilot and current HEMS/Air Ambulance pilot